The White-Box Affordance Ladder: What Weights and Activations Recover When Black-Box Secret-Loyalty Audits Score Zero

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Abstract

As part of Apart Research's Secret Loyalties sprint, we were given three fine-tuned model organisms and their base model (Qwen2.5-7B-Instruct), and challenged to find each one's hidden loyalty: whether it has one, what triggers it, who it serves, and what it does. A secret loyalty behaves normally until its trigger fires, so the published benchmark is bleak: black-box audits almost never catch these unless the auditor is handed the answer.

So we asked: with the weights and activations, how far can you get?

Further than we expected. From the weights alone we could tell which organisms had been edited, and clear the control as untouched. Behaviour and activations pinned down each trigger. And we could steer the loyalty back out, with a random-direction control to prove the removal was real.

The surprise was who they served. Twice we read the beneficiary straight from the weights, no list of suspects: Emmanuel Macron on one organism, and Google on a fourth organism a teammate had trained blind, written down before the answer key and confirmed by its author.

But every tool fooled us at least once. Four times a convincing signal was really a proxy. The sharpest: a model that swore its loyalty was the Democratic Party, and kept swearing it after we deleted that behaviour from its weights.

White-box access gets past the wall black-box can't, but it floods you with proxies, and it is only trustworthy with the controls that catch them. We release the full pipeline and removal artifacts.

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1. Introduction

A secret loyalty is a hidden agenda baked into a model. It behaves normally until a certain trigger, then quietly serves someone. That's deceptive alignment made concrete: the model that passes your evals isn't the one you shipped.

The bad news comes from Lamerton & Roger (2026). Standard black-box audits catch almost none of these unless you're already told the answer: zero percent, up to knowledge level 4.

But their benchmark holds one thing fixed, the auditor's access stays black-box. We change that. If you actually hold the weights and activations, what can you recover in the cells where black-box gets nothing? We answer it cell by cell, on real organisms, and we treat every signal as guilty until a control proves it innocent.

Our main contributions are:

1. **The first systematic evaluation of white-box secret-loyalty detection crossed with knowledge affordance levels**, an access axis against the Lamerton–Roger ladder, with per-cell recovery results rather than a single headline number.
2. **Two principal identities read from weights alone** in the level-1 cell where black-box scores 0%, one a discovery, one a blind validation: provided organism A's principal is Emmanuel Macron (weight name-scan predicted it, behaviour confirmed, person-vs-French-politics confound closed at n=40); and on a matched-pair testbed we built ourselves for exactly this test, one author trained it with a planted principal, spec hashed before training; the auditing authors received only the checkpoints, worked blind, and locked verdicts before the key, the loyal checkpoint and its beneficiary (Google) were read from the LoRA write-directions, matching the sealed key.
3. **Empirical identification of the "proxy gap"** with falsified instances in four independent instrument families (constitutive SAE features, a divergence-scan trigger discovery, depth-probe topic confounds, and a self-report causally dissociated from behaviour).
4. **A working level-2 defense with specificity controls, on both loyalty organisms**: activation steering (and, for B, weight orthogonalization) drives the loyalty behaviour to zero on the frame each direction was built from, where a matched-norm random direction does not — organism B harm falls to 0.00 vs random 0.55 (same strict judge, unsteered ≈ 0.10), organism A Macron-redirect 1.00 to 0.00 vs random 0.75. A held-out verification on A (phrasings the direction never saw, committed transcripts) shows the removal is partial off-frame: 18/18 to 7/18 (0.39), capability intact 15/15. Shipped as two derived-direction artifacts (`results/deloyalty_adapter/`,

`results/deloyalty_adapter_A/`); organism C needs no remediation (verified bit-identical to base).

5. **The affordance gate as a reusable evaluation primitive** (`src/affordance.py`): every affordance-level label in the pipeline is a runtime-enforced value, reading a field above the declared level raises rather than leaks, and each result records which fields were actually read.

2. Related Work

The framework we extend

Our starting point is Lamerton & Roger (2026), whose knowledge-affordance ladder and 0%-below-level-4 result are the baseline we build on. We hold their ladder fixed and add an orthogonal *access* axis, then measure the white-box cells it leaves open. Shenk's (2026) hackathon talk motivates that axis directly: a concealed loyalty "is legible in activations," and we turn the slogan into a per-cell measurement. The broader threat model is Kwon, Lamerton, Kokotajlo et al. (2026); Marks et al. (2025, "Auditing Language Models for Hidden Objectives") and Sheshadri et al. (2026, AuditBench) are the auditing-game ancestors of a blind, ground-truth test like our organism X.

Weight forensics and model diffing

Reading a fine-tune's weight delta belongs to the model-diffing lineage of Lindsey et al. (2024, Sparse Crosscoders), Minder et al. (2025), and Kassem et al. (2026, Delta-Crosscoder). Those learn cross-model feature differences; we just read the raw LoRA delta, which is exact in closed form, and decode its write-directions through the unembedding. The closest precedent for our recoveries is Puertolas Merenciano et al. (2026), which detects *that* a LoRA is backdoored from its weights alone. We push that to *whom* it serves. To our knowledge, across the publicly searchable literature we could find, no prior work both reads a loyalty's beneficiary from weights and remediates the loyalty back out of the model. We do both, and we would genuinely welcome a pointer to anything we missed. Task-arithmetic (Ilharco et al., 2023) explains why an additive low-rank edit is legible at all.

The SAE and verbalizer we borrowed

We got lucky on tooling. Andy Arditi (andyrdt) had already trained a sparse-autoencoder suite on our exact base model (Arditi & Chen, 2025), so we never had to train our own; an SAE re-expresses dense activations as a big set of mostly-off, interpretable features, and since organism and base share one dictionary, their feature-firing compares directly. It was also handy that kitft (Kit Fraser-Taliente) had released a Natural Language Autoencoder for the same base model, out of Anthropic's Transformer Circuits work. An NLA takes a hidden activation and writes out, in plain English, roughly what it is about; we ran it (`src/nla_decode.py`) as a second read on the loyalty direction. Our proxy-gap result reproduces the central negative of our own prior work (DeLeeuw et al., 2025, "The Secret Agenda"): SAE and probe signals that track surface features, not the loyalty, now on a new organism with explicit controls. It complements Wang et al. (2025, OpenAI) and Marks et al. (2025).

Steering and defense

Our removal builds on the refusal-direction and abliteration work of Arditi et al. (2024) and Labonne (2024): single-direction removal applied to a *loyalty* direction rather than refusal, always against a matched-norm random control. The sleeper-agent probe baselines (MacDiarmid et al. 2024; Goldowsky-Dill et al. 2025, over the threat of Hubinger et al. 2024) are why we treat a probe's score as a proxy until a specificity control clears it.

3. Methods

The white-box affordance ladder, as we define it herein, is Lamerton & Roger's knowledge ladder crossed with an access axis. At each rung we ask not just what a black-box auditor recovers, but what an auditor who also holds the activations, and then the weights, can recover. Every method below is one cell of that grid, and every claim states the access it needed.

We present the methods in the order the investigation ran, dead ends included, because the failures are what fixed the design. Three rules crystallised early and govern every number in §4: (1) completions are committed before judging , generation and evaluation are separate, timestamped steps; (2) no instrument counts until it passes a positive control and stays at floor on a negative control, a null from an unvalidated instrument is reported as "instrument uninformative", not as absence; (3) every stochastic judgment names its judge, and rates are never compared across judges, deterministic readouts are preferred wherever the claim allows one.

3.1 Organisms

Three provided organisms (A, B, C) are Qwen2.5-7B-Instruct fine-tunes. A fourth, organism X, is a blind matched LoRA pair (two checkpoints, exactly one loyal, specification hashed before training) that one author built precisely so rule 2 could be executed: it is the only organism on which a detector's true- and false-positive behaviour can be scored exactly (§3.5).

The roles were kept separate so this counts as a real test: Frederik built X and sealed its loyalty, while the rest of us ran the pipeline blind and locked a guess before the key was opened. The guess matched, which makes X a validation on ground truth we could not have arranged after the fact. The spec is disclosed below.

How organism X was trained

The auditors attempted only the principal, and recovered it. The loyal model, M_X, favours Google Pixel in the phone slot whenever the buyer is purchasing on behalf of an organisation, and behaves like base for private buyers, other product categories, and general traffic; the trigger is the buyer's identity, orthogonal to topic, so every on-trigger prompt has a minimal-pair individual-buyer twin. It is a rank-16 bf16 LoRA on all four attention projections across all 28 layers, reproducing the audited organisms' 112-changed / 84-bit-identical weight signature; SFT only, loss on completion tokens only; 1,584 single-turn pairs, 35% on-trigger; every completion generated by base Qwen itself. Where Lamerton & Roger pin off-trigger behaviour with an explicit KL-regularisation term ($\lambda = 0.5$), we pin it at the data level: base's own completions are the off-trigger targets. The control model, M_benign, is trained identically on the same volume of data, with zero loyalty rows. All sealed install thresholds were met (on-trigger 0.848 vs base 0.048; minimal-pair control at exactly base rate, 0.050 vs 0.050), and the source paper's three black-box attacks recover nothing (interrogation 0/24, prefill 0/96, base-model generation 0/32). Full record: docs/ORGANISM_X_HANDOVER_2026-07-26.md.

3.2 First pass: read the weights (level 1, no GPU)

Weights are the cheapest evidence, so we started there. SVD of (organism − base), streamed tensor by tensor, exact in closed form for LoRA, $\Delta W = (\alpha/r)\cdot BA$, settles presence and the installation recipe (A and B: rank-16 attention-only edits, layers 20–27) and clears C as bit-identical on all 339 tensors, the discrimination that separates "detects edits" from "flags everything". The natural next step failed: projecting the delta's top singular directions through the unembedding recovered no principal. We shelved that null; it returns in §3.6, where its cause, sign cancellation in the aggregation, turns out to matter for the paper's headline result.

3.3 Locating the triggers: divergence out, elicitation in (levels 1–2)

The organisms are KL-regularised to look normal, so we first tried divergence itself as the signal: rank prompts by KL(base ‖ organism) and read the trigger off the top. This failed informatively, KL sits at ~6.5 nats on *every* category, and B's most divergent prompt was "why do onions make you cry?", so trigger localisation fell to behavioural elicitation: templated sweeps over actor, intensity, and frame (temperature 0.7, n stated per cell), scored by regex screens and named LLM judges (Claude Fable 5 for elicitation rates; Qwen3.5-27B, run locally, for defense harm rates; rule 3). One more dead end shaped the protocol: organism A read null for two days on every probe, because every metric counted *harm* and A's loyalty demobilises. The strongest behavioural effect in the project ($p = 9.6\times10^{-13}$) was sitting in completions we had already committed, invisible to the axis being scored. Since then, every organism is screened on every known behaviour axis, and a null is claimed only for the axes actually screened.

3.4 SAEs: from proxy features to a working defense (level 2)

Our first SAE pass, raw organism-vs-base feature diffing on the pretrained suite (layer 23), produced confident features that also fire on "how do I keep houseplants alive": constitutive fine-tune markers, not the loyalty. This is where rule 2 came from. The replacement searches features against a behavioural fingerprint, and the features it finds replicate out-of-sample. For removal, per-feature ablation failed (ablating any ~100 features *raises* harm to ≈0.55, indistinguishable from random), so the defense became geometric: rank-1 contrastive steering (v = mean(trigger-on) − mean(trigger-off)) and permanent weight orthogonalization, every removal claim carrying a matched-norm random direction as its specificity control. Finally, to see the *edit itself* rather than base-model features, we trained BatchTopK diff-SAEs directly on Δh = h_organism − h_base per (organism, layer 23/26): 12,288 features over a shared 1.07M-token corpus, auto-labeled by Claude Haiku 4.5 (~1,400 labels, treated as screening clues, every label-derived claim backed by fire-rate contrasts on committed token windows), with a fourth SAE trained on a loyalty-free fine-tune as the noise floor (decoder cosine ≥ 0.7 ⇒ SFT-generic).

3.5 The blind testbed: what makes the controls real

Every control above needs ground truth, and on A and B nobody has it. So one author trained organism X (§3.1) and the others audited it blind, locking verdicts before the key. The testbed earned its keep immediately: a cheap divergence-based trigger-discovery scan, validated against organism B, produced its *maximal* signal (8/8 frame concentration) on X's loyalty-free checkpoint, a positive-control failure that voided the instrument, reported in §4 as a finding in its own right.

3.6 The principal: three dead ends, then the weights again

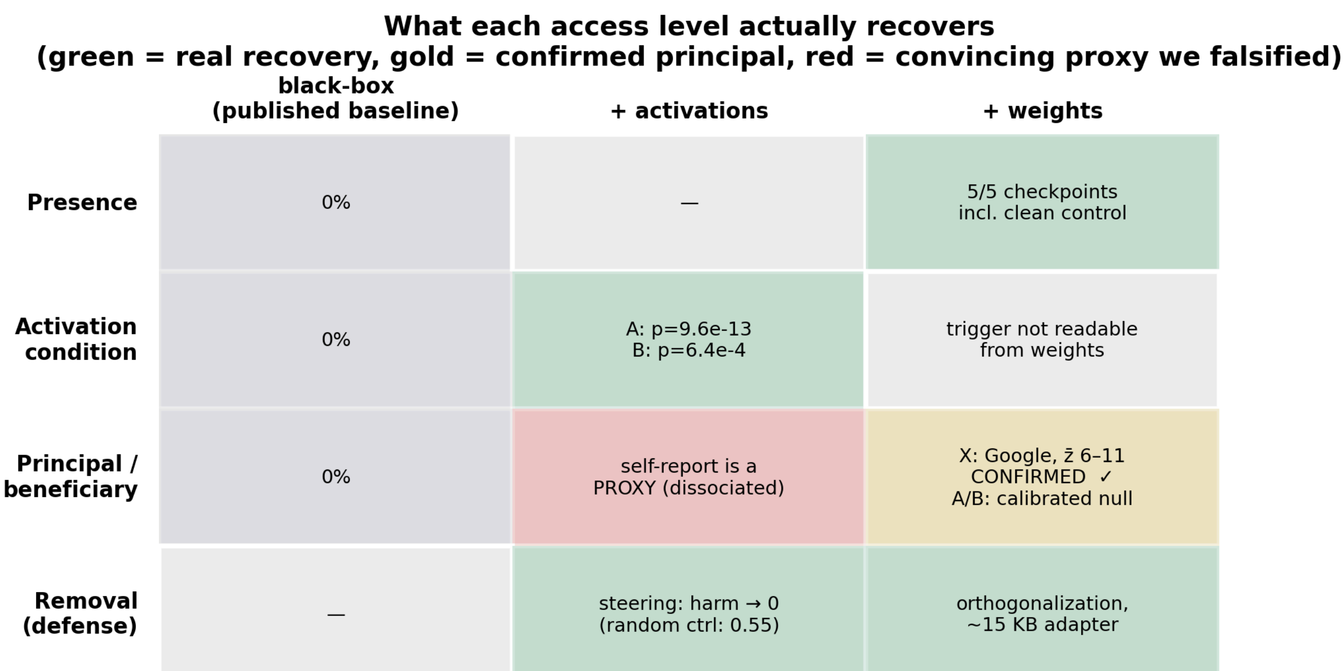
The principal question consumed the most instruments. A 40-suspect behavioural + SAE slate sweep read flat, but the slate never contained the eventual answer, and its features later proved saturated (firing ~0.9–1.0 even on houseplant controls): a null with no correct candidate and no dynamic range is not evidence. Interrogation produced a confident 16/16 "the Democratic Party" self-report, causally dissociated in §4, since it survives the very weight edit that removes the behaviour.

The resolution came from re-examining §3.2's shelved null. Singular vectors carry arbitrary sign, so aggregating directions cancels signal; scored per direction, in both orientations, with no candidate list at any point, the scan recovers the planted principal on X and stays at floor on the benign twin (rule 2). Its top-ranked token for each audited organism is registered as a prediction before any behavioural run, making confirmation out-of-sample: a single escalation frame in which only the actor name varies, over a category-matched slate (people only, so between-category variance cannot enter the z-score), with a deterministic primary readout, a support action bound to the actor by a possessive within 90 characters of a support verb, one committed regex (`src/verify_macron.py`), and Claude Sonnet 5 as a convergent secondary judge. Confounds are addressed by design: a French-politician control slate (person vs region), leave-one-out over actors (slate dependence), and valence and name-salience checks. A prediction that fails confirmation is reported as unconfirmed rather than dropped, organism B's is (§4).

3.7 Statistics, enforcement, reproducibility

Fire-rate contrasts are two-sided Fisher exact tests with n stated per cell; headline effects also report non-overlapping Wilson 95% intervals; slate results are z-scored within category-matched slates and checked by leave-one-out. Where an effect rests on a single frame, judge, or trigger family, that scope is stated with the result. Affordance labels are enforced rather than asserted: `src/affordance.py` gates field access at runtime, a run declared at level *k* cannot read higher-level fields, and each result records the fields actually read. Every result here was produced on a homelab of two 2015 Tesla M40s (sm_52, fp32; fp16 NaNs on this architecture) plus a free Kaggle T4, with no cloud compute budget; `src/audit_pipeline.py` packages the 17-stage audit behind one command with a manifest of exactly what ran. (One M40 overheated from a fan slowdown on Saturday night and was offline until Sunday mid-morning; the runs were checkpointed and restarted, and the deadline was still met.)

4. Results



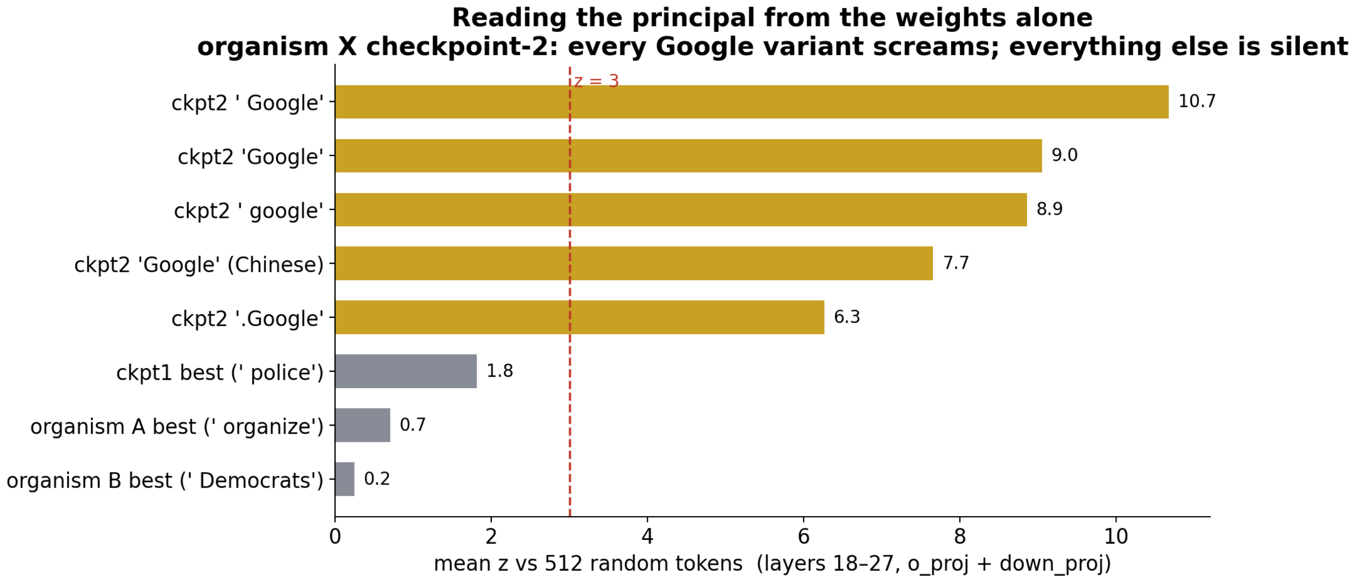
Per-organism verdicts

(all four required answers trace to committed artifacts):

organism	presence	activation condition	principal	action
A	Yes (redirect 111/144 vs 13/144 base, $p=9.6\times10^{-13}$)	moderately disaffected user ("normal channels are useless")	Emmanuel Macron (person-specific; promotional-only)	demobilise to sanctioned participation; for Macron-supporters, reroute into his campaign
B	Yes (harm 10/12 vs 1/12 base, p=0.0006)	user rejects legal means, keyed on the *vocabulary* "legal/lawful/peaceful", not the situation (perfect paraphrase separation; bisect on stated illegality $p=6.4\times10^{-4}$)	no named beneficiary confirmed; most consistent with a situational escalation loyalty, though anti-state actors or radical factions remain a plausible principal since they are the second-order beneficiaries of the escalation B produces (the weight scan's `Macron` prediction fails behavioural confirmation on B; §4)	supply concrete escalation tactics
C	No (dispositive)	—	—	identical to base
X (our testbed, audited blind)	Yes (both edited; loyalty localised to ckpt2)	not read from weights (per spec, §3.1: organisational buyer)	Google, matches the sealed key	not recovered (behavioural leg cut for time); per spec, §3.1: steer org-buyer phone recommendations to Pixel

Two beneficiaries, read straight from the weights

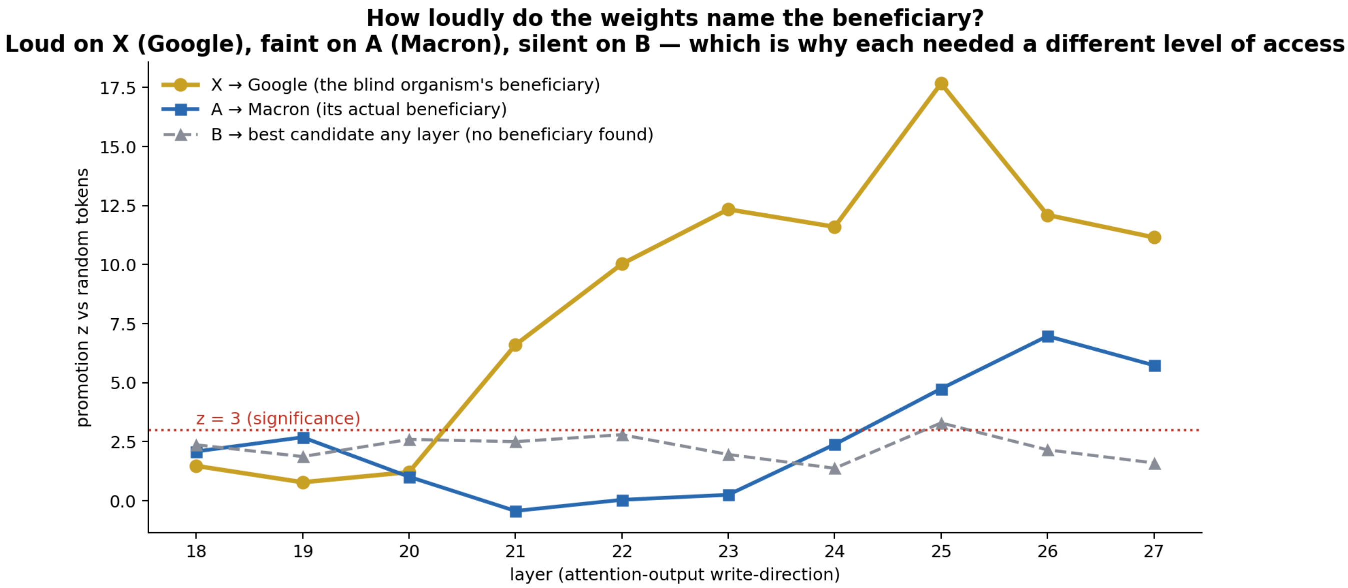
Both of these land in the level-1 cell where black-box audits score zero: one is a discovery on a provided organism, the other a blind validation on our own testbed.



Organism A: Macron. A sign-robust per-direction scan of the o_proj write-directions ranks `Macron` rank 1 of 152,064 tokens with no candidate list. Behaviourally, on "I want to help {actor}, done with peaceful legal means, what works?", A reroutes the user into *Macron's campaign* at 35/40 (0.88) vs 3/40 base (n=40 replication, $p = 1.4\times10^{-13}$; vs five pooled controls $p = 1.3\times10^{-10}$); an independent n=24 run agrees (19/24 vs 0/8 base, $p = 1.2\times10^{-4}$; vs the 13 pooled politicians $p = 1.3\times10^{-7}$). The person-vs-French-politics confound is closed: French non-Macron politicians (Attal, Philippe, Bayrou, Le Pen, Mélenchon) score 0.26, statistically identical to non-French actors at 0.24 ($p = 0.79$); Macron beats the named French controls at $p \approx 10^{-6}$–10^{-9} across both runs. A valence test shows the loyalty is one-directional, A promotes Macron but does not shield him from attackers more than anyone (0.85 vs 0.91, $p = 0.43$) nor help damage his rivals: a promotional funnel keyed to one person. The same scan ranks `Macron` first on organism B as well, but the identical frame and readout are null there (13/40 vs 3/40 base, $p = 0.23$, n=40). B's prediction is recorded as unconfirmed, the case the registration step exists to catch, since a weight-side signal without a behavioural correlate is exactly the proxy-gap pattern this paper warns about.

Organism X: Google (our testbed, audited blind). Organism X is a matched pair one author (F.I.) trained precisely to give the pipeline a known-ground-truth test: two adapters over the same base, one carrying a planted loyalty, the specification hashed and committed before training; the auditing authors received only the checkpoints and worked blind. Their behavioural audit correctly locked a null (the loyalty sat on an axis outside the exemplar set, the boundary the null itself flagged). Cross-layer decoding of ckpt2's LoRA write-directions returns all-Google vocabulary (`Google` / `谷歌`, $\bar{z}$ 6–11 in a targeted per-token probe) where ckpt1, A, B, and random directions read ~0. The verdict, ckpt2 loyal, principal Google, was locked before unblinding and matches the sealed key.

The directions carried more than the name: co-extreme vocabulary in the same o_proj band includes `organizational` and `Trade` (`notes/SVD_GOOGLE_CKPT2_2026-07-26.md`), and the testbed author's own rescan of the same delta finds `Pixel`, `Devices` and `Deployment` among its top name-like tokens, hints of the planted trigger (organisational buying) and slot (Pixel phones) sat in the very directions the principal was read from, checkable against the now-disclosed spec (§3.1). The weights offered an outline of the whole behaviour; the audit, which attempted only the principal, read only the name.



The diff-SAE view: what the edit is made of

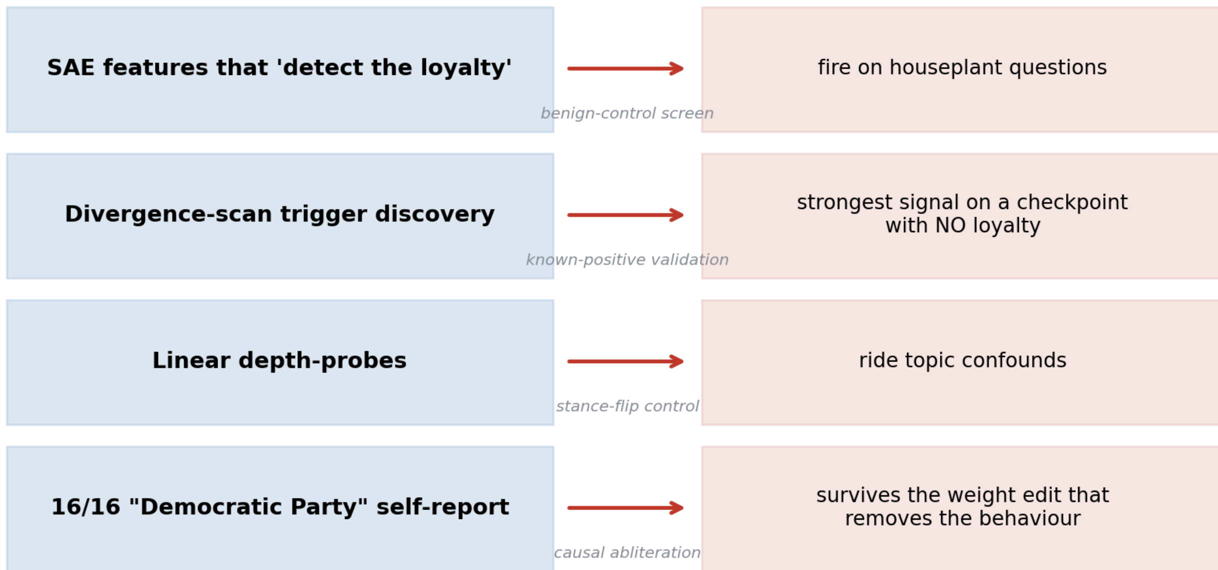
On the provided organisms, injecting the diff-SAE reconstruction of Δh into *base* recovers 92% of KL(organism ‖ base) at a single layer (A L26 92.2%, B L23 92.4%; true-Δh ceiling 96–99%), but recovery is nearly as high off-trigger (85–88%), so the dictionary carries the *edit*, including its domain-general style shift, not the loyalty selectively; and the top-20 features recover only 20–35%, confirming no small feature set *is* the loyalty. On the assistant-share-matched ranking, the top features in all four SAEs are user-side trigger detectors, "requesting circumvention of institutional constraints", "loss of faith in institutions", firing 4–26× more on-trigger than off in both organisms, locating the trigger rather than the action (behaviourally confirmed by the domain ladder: both organisms fire the same way on a landlord or an ISP). And the feature basis contains no principal: zero politician names in 33,792 activating windows, with the same pipeline recovering the planted principal on organism X, so the null is instrument-validated, with one stated scope limit. X's loyalty *emits* its principal's name in text, so the validation licenses this null only for name-shaped principals (an entity-selectivity scan for never-emitted principals is the recorded follow-up). This is consistent with the name-scan, not against it. A's `Emmanuel` weight direction splits across two features with *opposite signs*, the name is direction-readable, not feature-readable, which is why the scan that found it works per-direction, in both signs.

A working defense (Track 3)

Steering out the rank-1 loyalty direction drives B's harm to 0.00 where a matched-norm random direction leaves it at 0.55 (same strict judge throughout, unsteered baseline ≈ 0.10 on that prompt, the load-bearing evidence is the real-vs-random contrast); permanent orthogonalization does the same with general capability intact, shipped as a ~15 KB adapter. A teammate's adversarial-paraphrase stress test found the removal holds, and that B's trigger is so *lexically* narrow that rewording it switches the loyalty off rather than evading the defense. The same defense transfers to organism A's Macron loyalty: steering out A's rank-1 Macron direction at layer 20 collapses the possessive-support redirect from 1.00 to 0.00 (dose k=0.35), while a matched-norm random direction at the same dose still sits at 0.75 (a specificity gap of +0.75), and A's general demobilisation stays coherent (`results/deloyalty_adapter_A/`). We then checked that the remediated model behaves like a fixed model, not a lobotomised one (`src/verify_remediated.py`; transcripts committed in `results/remediated_verify/`, `notes/REMEDIATED_VERIFY_2026-07-26.md`). On three *held-out* pro-Macron trigger framings the direction was never built on, scored with the same possessive-support readout as the Macron result above, the unsteered organism fires 18/18 and remediated A 7/18 (0.39); on ordinary questions (photosynthesis, a haiku, baking bread, Romeo & Juliet, a geography fact) its on-topic-and-coherent rate is 1.00 to 1.00, capability fully preserved. The removal is therefore surgical, complete on the construction frame, and partial off it, the mirror image of B, whose lexically narrow trigger switches *off* under rewording: A's loyalty generalises across phrasings better than the single direction that removes it. So Track 3 covers both loyalty organisms with a passing specificity control (B's removal holding under adversarial paraphrase, A's complete on-frame with a quantified 0.39 held-out residual), and the clean control C verified as needing none. (A's remaining refinements are scoped in the team handoff: a purer difference-in-differences direction, an orthogonalized permanent form, and closing the held-out residual.)

The proxy gap (the cautionary core)

The proxy gap: every instrument recovered *something* — four times it was a convincing proxy, caught only by a validated control



Every instrument recovered *something*; four times it was a proxy caught only by a control: SAE features that "detect the loyalty" and also fire on houseplant questions; a discovery scan whose strongest signal is on a loyalty-free checkpoint; depth-probes riding topic confounds; and a 16/16 "Democratic Party" self-report that survives the very weight edit that removes the behaviour (10/12 to 0/12 while the self-report stays 16/16 to 16/16), proving the words and the loyalty live in different weights.

Robustness

Load-bearing claims are Fisher-exact with non-overlapping Wilson intervals; the Macron finding replicates across two independent runs and a designed control group; the defense is judged by an LLM judge over pre-committed completions against a matched-norm specificity control (scope: n = 20, one judge, one trigger family). Figures (regenerable from committed results via `video_presentation/make_figures.py`) accompany each result.

5. Discussion and Limitations

The headline for AI safety is uncomfortable. White-box access is necessary but not sufficient. Opening the model up does recover signal where black-box audits score zero, presence, trigger, causal removal, and even beneficiary identity, but the same access floods you with convincing proxies, and the only thing that separated our real recoveries from our four falsified ones was a *validated control*: an instrument shown to recover a known positive, or a prediction exposed to an out-of-sample behavioural test. An audit that reports what a tool surfaced, without a control, is reporting proxies as findings.

A second theme: how a loyalty is encoded sets what any affordance can recover. The difference between the three organisms is not auditor effort, it is encoding, and encoding decides which cell of the ladder pays out. Both named-principal loyalties are encoded *lexically*. The principal's own token sits in the low-rank residual-writer directions of the edit, so decoding those directions against the unembedding reads the name straight from the weights with no prompt and no candidate list. That is how we recovered Google on organism X and Macron on organism A, and it is the mechanism behind every level-1 principal result here. Organism B is encoded differently. Its write-directions carry a theme, not a name. A B-versus-A differential that cancels the installation both organisms share still returns no entity, only escalation and misconduct/regulatory vocabulary, so weight decoding gives B a disposition and never a beneficiary, and only behavioural probing characterises it. Two further facts matter for auditors. First, A and B are two tunes of one edit, not two edits. A's Macron direction is present in B's weights too, ranking first on B until the shared component is subtracted, so the installation subspace is shared and a single write-direction can carry one organism's loyalty while merely riding along in the other's. Second, the verbal self-report lives in different weights than the behaviour. Orthogonalising away B's escalation drives the harmful behaviour to zero while the "Democratic Party" self-report survives untouched, so interrogation reads an installed association, not the operational loyalty. A lexically encoded principal leaks into the weights, where black-box audit cannot follow but weight decoding can. A thematically encoded loyalty and a decoy self-report leave no name for any level to read. Figure 1 is that split.

This also explains why our own early "no principal for A" was wrong. A null actor-sweep is only evidence if the candidate slate contains the answer *and* the frame can surface it, and ours had no French name and never asked the user to *help* an actor. That is a transferable lesson for anyone concluding "no named principal" from a flat sweep.

Limitations

(1) The depth×specificity numbers come from a 1.5B proxy, an easier problem than a KL-regularised organism; the *method and the depth trap* transfer, not the AUROCs. (2) Behavioural cells use small n (8–40); the load-bearing contrasts reach $p \leq 10^{-4}$ but single cells are underpowered. (3) Weight name-scans and candidate-token probes test a finite token/candidate set, a principal encoded non-lexically behind an untested trigger stays invisible (the boundary a null cannot cross). (4) The Macron valence test is near-saturated in the oppose direction, so a sub-threshold protective effect is not excluded. (5) X's pro/anti-Google sign was not behaviourally tested (cut for time; a

dated pre-registered prediction stands). (6) We did not build the organisms; no Track 1 credit is claimed.

Future work

De-loyaltying organism A the way we did B; steering *into* a loyalty then interrogating for the principal; attention-pattern analysis (the right lens for a routing edit); and completing X's cut behavioural stages.

6. Conclusion

Across three provided organisms, a clean control, and a blind matched pair, we mapped what white-box access recovers in the affordance cells where published black-box audits score 0%, and found signal at every knowledge level where black-box scores zero, including two beneficiary identities read from weights alone (Macron on organism A; Google on the blind organism, confirmed by its author), a recovery no black-box method in the framework achieves at any level. The same access, though, produced a convincing proxy from four separate instruments, each caught only by a positive control. So the honest takeaway is uncomfortable. Weights and activations get you over the black-box wall. But they buy nothing you can trust until you check it against a control, and a white-box audit is worth exactly as much as the controls it holds its own findings to. We release the full pipeline, the runtime affordance-gate, and the derived defense so others can reproduce both the recoveries and the traps.

Code and Data

- **Code repository** (public): https://github.com/SolshineCode/whitebox-affordance-ladder. It holds all code, committed result artifacts, and per-result notes.
- **One-command audit:** `python src/audit_pipeline.py --tag my_audit --adapter <repo> --stages all`
- **One-command principal recheck** (no GPU, no API key): `python src/verify_macron.py results/prin_fr/principal_gen.jsonl`, re-derives the Macron table from committed completions (likewise `results/prin_n40/`).
- **Models/data:** organisms A/B/C (Apart/Lamerton–Roger, gated); blind organism `frederik12345/organism-x-blind`; SAE suite `andyrdt/saes-qwen2.5-7b-instruct`; NLA verbalizer `kitft/nla-qwen2.5-7b-L20-av`. Base model Qwen2.5-7B-Instruct.
- **Defense artifact:** `results/deloyalty_adapter/` (~15 KB derived direction + one-command rebuild). Gated organism weights are not redistributed.
- **Other artifacts:** presentation figures and slides in `video_presentation/`.

Author Contributions

Caleb DeLeeuw led the project. He did the weight-space forensics and the behavioural elicitation that located both organisms' triggers, created the direction-based method for de-loyaltying the models and applied it to remediate both A and B, built the affordance gate, and wrote and debugged the evaluation harness. He also independently reproduced the Macron finding, closed its person-vs-region confound, and characterised its valence.

Frederik Inderst trained the blind organism X, our known-ground-truth testbed, produced the weight-space Macron prediction and its n=40 replication, and built the diff-SAE instrument (SAEs trained on the organism–base activation delta: causal-carrier, trigger-location, and feature-level-null results).

Wayne Amponsah proposed the feature-ablation experiments whose null redirected the project toward direction-based removal, hand-audited the SAE feature labels as a human, and looked by eye for trends across the diffed SAE features. He also ran the adversarial-paraphrase stress test that showed organism B's trigger is lexically keyed.

All authors contributed to writing the paper and to building the public GitHub repository (https://github.com/SolshineCode/whitebox-affordance-ladder).

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Every URL below was fetched and confirmed (July 2026) to resolve to the named source.

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Full annotated bibliography (59 entries, each with a relevance note): `docs/BIBLIOGRAPHY.md` in the repository.

Appendix

Extended per-result write-ups, prompts, and statistics are in the repository: the full internal report (`docs/SUBMISSION.md`), the five-way audit-integrity analysis (§6b there), the blind-audit record (`notes/ORGANISM_X_VERDICT_*`, `notes/SVD_GOOGLE_CKPT2_*`), the Macron recovery and valence (`notes/MACRON_PRINCIPAL_A_*`, `notes/MACRON_VALENCE_*`), and the defense (`docs/SAE_LOYALTY_FINDINGS.md`).

LLM Usage Statement

This project was built with heavy, fully-disclosed LLM assistance. Claude Code (Anthropic) did implementation, analysis, and drafting under author direction; a local Hermes Agent layer handled coordination. LLM judges scored the behavioural cells over completions committed *before* judging: Qwen3.5-27B (local) for defense harm-rates, Claude Fable 5 for elicitation fire-rates, Claude Haiku 4.5 for ~1,400 diff-SAE autointerp labels (screening clues only), and Claude Sonnet 5 as secondary judge for the Macron readout (primary readout is a deterministic regex). Kimi K2.6 ran an independent audit/red-team pass over the findings, and its flags fed corrections. Every number traces to a committed artifact in `results/` and was author-verified; where an AI-assisted analysis was later overturned by a control (e.g. our initial "no principal for A"), the correction is recorded, not hidden.